

Black-Box Test Plan (Simple, General CS3IOS Template):

Test Case Scenario	Test the written function which take integer value n as input greater than 0 and less than 10. print all the factors of it separated by space.
Test Technique Chosen	Simple Boundary Value Analysis
Reason for Test Technique Chosen	The range of input value n is given

Code under Black-Box Testing (for marker's reference)

```
def printFactor(n):  
    for i in range(1,n):  
        if n%i==0:  
            print(i,end=" ")  
    print()
```

Valid input range / set (for marker's reference)

1 - 9

Test Case Calculations (if appropriate)

Since you selected Simple Boundary Value Analysis as the testing technique

$$\begin{aligned}\text{Total_number_of_test_cases} &= 4 * \text{number_of_input} + 1 \\ &= 4*1+1 = 5\end{aligned}$$

Test cases:

- Extremes: (maximum value, minimum value)
- Near extremes: (maximum value -1, minimum value +1)
- Normal: midpoint value (maximum – minimum value)/2)

$$n_max = 9$$

$$n_min = 1$$

$$n_max - 1 = 8$$

$$n_min + 1 = 2$$

$$n_nominal = 4$$

Test Case ID	Pre-Conditions	Test Step Details	Input(s) (Test Data)	Expected Result	Actual Result	Pass/Fail
ID_1	n > 0 and n < 10	Call printFactor(n_max) assert Expected Result	9	1 3 9	1 3	Fail
ID_2	n > 0 and n < 10	Call printFactor(n_min) assert Expected Result	1	1	null	Fail
ID_3	n > 0 and n < 10	Call printFactor(n_max-1) assert Expected Result	8	1 2 4 8	1 2 4	Fail
ID_4	n > 0 and n < 10	Call printFactor(n_min+1) assert Expected Result	2	1 2	1	Fail
ID_5	n > 0 and n < 10	Call printFactor(n_nominal) assert Expected Result	4	1 2 4	1 2	Fail

Notes

When input set to 1, the function fails because of the for loop set as (1,1) and loop doesn't execute even once.

Given function has a small mistake in the loop range. It iterates from 1 to n-1 since range (1, n) excludes n, but you want the loop to include n as well, since n itself is a factor of n.

*Notes on this template:

- Template is recommended to be used in landscape orientation.
- You may add columns or additional sections if you want to.
- It is up to you how you title/divide each section of your testing documentation.
- Templates are usually company-generated based on the company's overarching testing strategy and policies, this is a very basic template that is designed to help you with CS3IOS.